

AIN ZARA ALUMINUM & GLASS

Tripoli Industrial Complex - Official Factory HQ & Engineering Directorate

COMPLETE ARCHITECTURAL SYSTEMS MASTER CATALOG

2026 EDITION | TENDER SPECIFICATIONS & FABRICATION MATRIX



ARCHITECTURAL PORTFOLIO & TECHNICAL MATRICES

Comprehensive engineering handbook covering 14 certified European extrusion systems, automated Lisec double/triple IGU glass processing, and Qualicoat powder coating standards.

CERTIFIED PRODUCT SERIES INCLUDED IN THIS MANUAL:

ARTOS SERIES :	SAT 120 Lift & Slide SLAT 64 Multi-Rail Sliding
HEKLA SERIES :	WAT 63 Thermal Break WA 55 Standard WA 45 Economic
BROMO SERIES :	CWA 50 SG Structural Glazing CWA 50 HV Concealed Vent
NEPAL SERIES :	FAT 70 Heavy-Duty Bi-Fold FAT 55 Panoramic Accordion
IDA & URAL :	IPA 45 / IPA 30 Office Partitions SA 65 Skylight & Roofs
LOGAN SERIES :	GSA 130 Automated Motorized Guillotine Vertical Systems

TRIPOLI INDUSTRIAL COMPLEX - MANUFACTURING DIRECTIVES

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CERTIFIED: ISO 9001:2015 * QUALICOAT CERTIFIED * QUALANOD ANODIZING * EN 12020-2



FACTORY INFRASTRUCTURE & AUTOMATED FABRICATION CAPABILITIES

- CNC Extrusion Processing: Dual-head automated precision miter saws (0.1mm tolerance), 4-axis profile machining centers.
- Polyamide Thermal Barrier Crimping: Automated knurling, thermal strip insertion, and multi-roller crimping to EN 14024.
- Automated Glass Processing: Lisec robotic double/triple insulating glass line with automated dual seal and 95%+ Argon gas fill.
- Horizontal Tempering Furnace: Convection-assisted tempering for up to 19mm safety glass conforming strictly to EN 12150-1.
- Surface Finishing Plant: Qualicoat-certified architectural electrostatic powder coating (60-80 microns) & Qualanod anodizing.
- Engineering & Tender Support: Dedicated Tripoli engineering desk offering cutting list optimization, shop drawings, and BIM files.

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Ain Zara
Glass Processing & Aluminum Manufacturing

SAT 120 - Lift & Slide Architectural System

SERIES: ARTOS SERIES | SYSTEM CODE: SAT120 | EXTRUSION: EN AW-6063 T6

ENGINEERING 2D CAD PROFILE SCHEMATIC



MECHANICAL & ARCHITECTURAL SPECIFICATION MATRIX		MANUFACTURING STANDARD
Base Aluminum Alloy	EN AW-6063 T6 (Extrusion to EN 12020-2)	
Frame Architectural Depth	120 mm (Dual Track) / 180 mm (Triple Track)	
Sash / Vent Profile Depth	50 mm Heavy-Duty Reinforced Sash	
Glazing Thickness Capacity	24 mm - 38 mm Double / Triple IGU	
Thermal Barrier Insulation	20 mm Polyamide PA66 GF25 Thermal Barrier	
Air Permeability (EN 12207)	Class 4 (EN 12207 / 600 Pa)	
Water Tightness (EN 12208)	Class 9A (EN 12208 / 600 Pa)	
Wind Load Resistance (EN 12210)	Class C4 (EN 12210 / 1600 Pa)	
Acoustic Sound Reduction	Rw = 37 - 42 dB	
Maximum Sash Weight Capacity	300 kg (Slide) / 400 kg (Lift & Slide)	

FABRICATION STANDARDS & QUALITY COMPLIANCE

- Extrusion Tolerances: Conforms strictly to EN 12020-2 / DIN 17615 high precision architectural standard.
- Surface Coating: Qualicoat Certified Electrostatic Powder Coating (60-80 microns) / Qualanod Anodizing (15-20 microns).
- Gaskets: EPDM weatherseals according to DIN 7863. Hardware groove: Standard European groove compatibility.
- Quality Management: Extruded and assembled under certified ISO 9001:2015 industrial manufacturing systems.

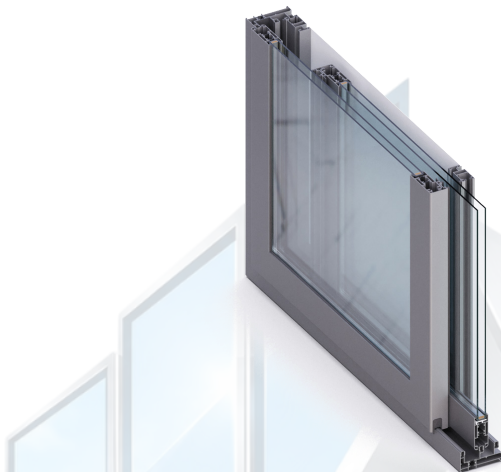
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SLAT 64 - Monorail & Multi-Rail Sliding System

SERIES: ARTOS SERIES | SYSTEM CODE: SLAT64 | EXTRUSION: EN AW-6063 T6

ENGINEERING 2D CAD PROFILE SCHEMATIC



MECHANICAL & ARCHITECTURAL SPECIFICATION MATRIX

MANUFACTURING STANDARD

Base Aluminum Alloy	EN AW-6063 T6 (Extrusion to EN 12020-2)
Frame Architectural Depth	64 mm (2-Rail) / 112 mm (3-Rail)
Sash / Vent Profile Depth	38 mm Curved / Interlocking Sash
Glazing Thickness Capacity	18 mm - 28 mm Insulated Glazing
Thermal Barrier Insulation	Standard Non-Thermal / Interlock Brush Gaskets
Air Permeability (EN 12207)	Class 3 (EN 12207 / 450 Pa)
Water Tightness (EN 12208)	Class 7A (EN 12208 / 300 Pa)
Wind Load Resistance (EN 12210)	Class C3 (EN 12210 / 1200 Pa)
Acoustic Sound Reduction	Rw = 32 - 36 dB
Maximum Sash Weight Capacity	160 kg per Sash

FABRICATION STANDARDS & QUALITY COMPLIANCE

- Extrusion Tolerances: Conforms strictly to EN 12020-2 / DIN 17615 high precision architectural standard.
- Surface Coating: Qualicoat Certified Electrostatic Powder Coating (60-80 microns) / Qualanod Anodizing (15-20 microns).
- Gaskets: EPDM weatherseals according to DIN 7863. Hardware groove: Standard European groove compatibility.
- Quality Management: Extruded and assembled under certified ISO 9001:2015 industrial manufacturing systems.

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WAT 63 - High-Performance Thermal Break Casement System

SERIES: HEKLA SERIES | SYSTEM CODE: WAT63 | EXTRUSION: EN AW-6063 T6

ENGINEERING 2D CAD PROFILE SCHEMATIC



MECHANICAL & ARCHITECTURAL SPECIFICATION MATRIX

MANUFACTURING STANDARD

Base Aluminum Alloy	EN AW-6063 T6 (Extrusion to EN 12020-2)
Frame Architectural Depth	63 mm Architectural Depth
Sash / Vent Profile Depth	71 mm Inward / Outward Rebate Sash
Glazing Thickness Capacity	20 mm - 42 mm High Acoustic IGU
Thermal Barrier Insulation	24 mm Polyamide PA66 GF25 Thermal Barrier
Air Permeability (EN 12207)	Class 4 (EN 12207 / 600 Pa)
Water Tightness (EN 12208)	Class 9A (EN 12208 / 600 Pa)
Wind Load Resistance (EN 12210)	Class C5 (EN 12210 / 2000 Pa)
Acoustic Sound Reduction	Rw = 38 - 44 dB
Maximum Sash Weight Capacity	180 kg (Tilt & Turn) / 220 kg (Door)

FABRICATION STANDARDS & QUALITY COMPLIANCE

- Extrusion Tolerances: Conforms strictly to EN 12020-2 / DIN 17615 high precision architectural standard.
- Surface Coating: Qualicoat Certified Electrostatic Powder Coating (60-80 microns) / Qualanod Anodizing (15-20 microns).
- Gaskets: EPDM weatherseals according to DIN 7863. Hardware groove: Standard European groove compatibility.
- Quality Management: Extruded and assembled under certified ISO 9001:2015 industrial manufacturing systems.

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WA 55 - Standard Casement Door & Window System

SERIES: HEKLA SERIES | SYSTEM CODE: WA55 | EXTRUSION: EN AW-6063 T6

ENGINEERING 2D CAD PROFILE SCHEMATIC



MECHANICAL & ARCHITECTURAL SPECIFICATION MATRIX		MANUFACTURING STANDARD
Base Aluminum Alloy	EN AW-6063 T6 (Extrusion to EN 12020-2)	
Frame Architectural Depth	55 mm Architectural Frame	
Sash / Vent Profile Depth	63 mm Overlap Sash with Drip Deflector	
Glazing Thickness Capacity	6 mm Monolithic - 32 mm Double IGU	
Thermal Barrier Insulation	Standard Hollow Non-Thermal Multi-Chamber	
Air Permeability (EN 12207)	Class 3 (EN 12207 / 450 Pa)	
Water Tightness (EN 12208)	Class 8A (EN 12208 / 450 Pa)	
Wind Load Resistance (EN 12210)	Class C3 (EN 12210 / 1200 Pa)	
Acoustic Sound Reduction	Rw = 32 - 37 dB	
Maximum Sash Weight Capacity	130 kg (Window) / 160 kg (Entrance Door)	

FABRICATION STANDARDS & QUALITY COMPLIANCE

- Extrusion Tolerances: Conforms strictly to EN 12020-2 / DIN 17615 high precision architectural standard.
- Surface Coating: Qualicoat Certified Electrostatic Powder Coating (60-80 microns) / Qualanod Anodizing (15-20 microns).
- Gaskets: EPDM weatherseals according to DIN 7863. Hardware groove: Standard European groove compatibility.
- Quality Management: Extruded and assembled under certified ISO 9001:2015 industrial manufacturing systems.

FACTORY DIRECTIVE: Official technical specifications verified for tender submissions, fabrication, and structural installation.

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WA 45 - Economic Casement & Fixed Window System

SERIES: HEKLA SERIES | SYSTEM CODE: WA45 | EXTRUSION: EN AW-6063 T6

ENGINEERING 2D CAD PROFILE SCHEMATIC



MECHANICAL & ARCHITECTURAL SPECIFICATION MATRIX

MANUFACTURING STANDARD

Base Aluminum Alloy	EN AW-6063 T6 (Extrusion to EN 12020-2)
Frame Architectural Depth	45 mm Compact Profile Depth
Sash / Vent Profile Depth	52 mm Casement Sash
Glazing Thickness Capacity	4 mm Monolithic - 24 mm Double Glazing
Thermal Barrier Insulation	Single Hollow Chamber Non-Thermal
Air Permeability (EN 12207)	Class 2 (EN 12207 / 300 Pa)
Water Tightness (EN 12208)	Class 6A (EN 12208 / 250 Pa)
Wind Load Resistance (EN 12210)	Class C2 (EN 12210 / 800 Pa)
Acoustic Sound Reduction	Rw = 30 - 34 dB
Maximum Sash Weight Capacity	90 kg (Window) / 120 kg (Door)

FABRICATION STANDARDS & QUALITY COMPLIANCE

- Extrusion Tolerances: Conforms strictly to EN 12020-2 / DIN 17615 high precision architectural standard.
- Surface Coating: Qualicoat Certified Electrostatic Powder Coating (60-80 microns) / Qualanod Anodizing (15-20 microns).
- Gaskets: EPDM weatherseals according to DIN 7863. Hardware groove: Standard European groove compatibility.
- Quality Management: Extruded and assembled under certified ISO 9001:2015 industrial manufacturing systems.

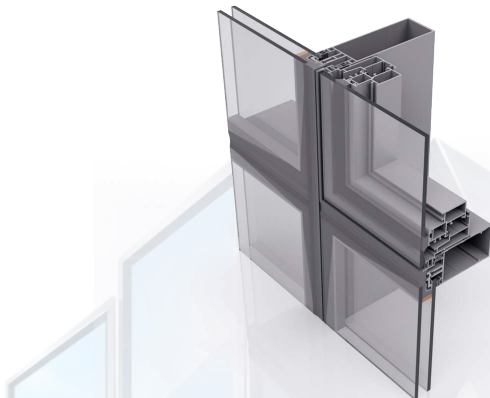
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CWA 50 SG - Full Structural Glazing Curtain Wall System

SERIES: BROMO SERIES | SYSTEM CODE: CWA50SG | EXTRUSION: EN AW-6063 T6

ENGINEERING 2D CAD PROFILE SCHEMATIC



MECHANICAL & ARCHITECTURAL SPECIFICATION MATRIX		MANUFACTURING STANDARD
Base Aluminum Alloy	EN AW-6063 T6 (Extrusion to EN 12020-2)	
Frame Architectural Depth	80 mm - 220 mm Structural Mullion Depths	
Sash / Vent Profile Depth	50 mm Exterior Sightline (Flush Glass)	
Glazing Thickness Capacity	24 mm - 44 mm Structural Silicone Bonded IGU	
Thermal Barrier Insulation	Continuous EPDM Multi-Fin Thermal Isolators	
Air Permeability (EN 12207)	Class AE 1200 (EN 12152 / 1200 Pa)	
Water Tightness (EN 12208)	Class RE 1500 (EN 12154 / 1500 Pa)	
Wind Load Resistance (EN 12210)	Design 2000 Pa / Safety 3000 Pa (EN 13116)	
Acoustic Sound Reduction	Rw = 42 - 48 dB	
Maximum Sash Weight Capacity	350 kg per Glazing Module	

FABRICATION STANDARDS & QUALITY COMPLIANCE

- Extrusion Tolerances: Conforms strictly to EN 12020-2 / DIN 17615 high precision architectural standard.
- Surface Coating: Qualicoat Certified Electrostatic Powder Coating (60-80 microns) / Qualanod Anodizing (15-20 microns).
- Gaskets: EPDM weatherseals according to DIN 7863. Hardware groove: Standard European groove compatibility.
- Quality Management: Extruded and assembled under certified ISO 9001:2015 industrial manufacturing systems.

FACTORY DIRECTIVE: Official technical specifications verified for tender submissions, fabrication, and structural installation.

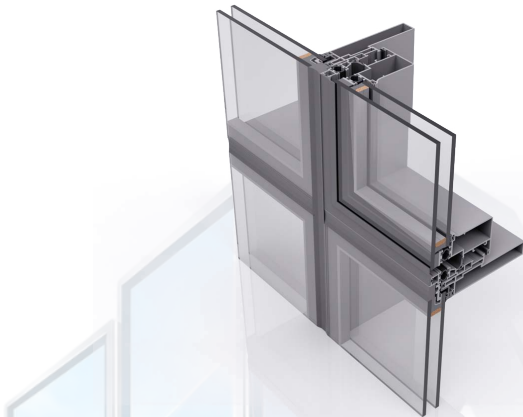
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CWA 50 HV - Semi-Structural Concealed Vent Facade System

SERIES: BROMO SERIES | SYSTEM CODE: CWA50HV | EXTRUSION: EN AW-6063 T6

ENGINEERING 2D CAD PROFILE SCHEMATIC



MECHANICAL & ARCHITECTURAL SPECIFICATION MATRIX		MANUFACTURING STANDARD
Base Aluminum Alloy	EN AW-6063 T6 (Extrusion to EN 12020-2)	
Frame Architectural Depth	80 mm - 220 mm Mullion & Transom Matrix	
Sash / Vent Profile Depth	50 mm Visible Grid with Hidden Operable Vents	
Glazing Thickness Capacity	24 mm - 44 mm Double / Triple Acoustic Glazing	
Thermal Barrier Insulation	Multi-stage EPDM Isolator with Polyamide Core	
Air Permeability (EN 12207)	Class AE 1200 (EN 12152 / 1200 Pa)	
Water Tightness (EN 12208)	Class RE 1200 (EN 12154 / 1200 Pa)	
Wind Load Resistance (EN 12210)	Design 2000 Pa / Safety 3000 Pa (EN 13116)	
Acoustic Sound Reduction	Rw = 40 - 46 dB	
Maximum Sash Weight Capacity	180 kg Concealed Parallel / Top-Hung Vent	

FABRICATION STANDARDS & QUALITY COMPLIANCE

- Extrusion Tolerances: Conforms strictly to EN 12020-2 / DIN 17615 high precision architectural standard.
- Surface Coating: Qualicoat Certified Electrostatic Powder Coating (60-80 microns) / Qualanod Anodizing (15-20 microns).
- Gaskets: EPDM weatherseals according to DIN 7863. Hardware groove: Standard European groove compatibility.
- Quality Management: Extruded and assembled under certified ISO 9001:2015 industrial manufacturing systems.

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FAT 70 - Heavy-Duty Thermal Break Bi-Fold Door System

SERIES: NEPAL SERIES | SYSTEM CODE: FAT70 | EXTRUSION: EN AW-6063 T6

ENGINEERING 2D CAD PROFILE SCHEMATIC



MECHANICAL & ARCHITECTURAL SPECIFICATION MATRIX

MANUFACTURING STANDARD

Base Aluminum Alloy	EN AW-6063 T6 (Extrusion to EN 12020-2)
Frame Architectural Depth	70 mm Architectural Frame
Sash / Vent Profile Depth	70 mm Heavy Flush Sash Assembly
Glazing Thickness Capacity	24 mm - 40 mm High-Performance IGU
Thermal Barrier Insulation	30 mm Polyamide PA66 GF25 Thermal Barrier
Air Permeability (EN 12207)	Class 4 (EN 12207 / 600 Pa)
Water Tightness (EN 12208)	Class 8A (EN 12208 / 450 Pa)
Wind Load Resistance (EN 12210)	Class C3 (EN 12210 / 1200 Pa)
Acoustic Sound Reduction	Rw = 36 - 41 dB
Maximum Sash Weight Capacity	140 kg per Folding Leaf (Up to 8 Leaves)

FABRICATION STANDARDS & QUALITY COMPLIANCE

- Extrusion Tolerances: Conforms strictly to EN 12020-2 / DIN 17615 high precision architectural standard.
- Surface Coating: Qualicoat Certified Electrostatic Powder Coating (60-80 microns) / Qualanod Anodizing (15-20 microns).
- Gaskets: EPDM weatherseals according to DIN 7863. Hardware groove: Standard European groove compatibility.
- Quality Management: Extruded and assembled under certified ISO 9001:2015 industrial manufacturing systems.

FACTORY DIRECTIVE: Official technical specifications verified for tender submissions, fabrication, and structural installation.

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FAT 55 - Panoramic Thermal Folding Door System

SERIES: NEPAL SERIES | SYSTEM CODE: FAT55 | EXTRUSION: EN AW-6063 T6

ENGINEERING 2D CAD PROFILE SCHEMATIC



MECHANICAL & ARCHITECTURAL SPECIFICATION MATRIX		MANUFACTURING STANDARD
Base Aluminum Alloy	EN AW-6063 T6 (Extrusion to EN 12020-2)	
Frame Architectural Depth	55 mm Architectural Frame	
Sash / Vent Profile Depth	55 mm Slim Folding Sash	
Glazing Thickness Capacity	18 mm - 32 mm Insulated Glass	
Thermal Barrier Insulation	16 mm Polyamide Thermal Barrier	
Air Permeability (EN 12207)	Class 3 (EN 12207 / 450 Pa)	
Water Tightness (EN 12208)	Class 7A (EN 12208 / 300 Pa)	
Wind Load Resistance (EN 12210)	Class C2 (EN 12210 / 800 Pa)	
Acoustic Sound Reduction	Rw = 33 - 38 dB	
Maximum Sash Weight Capacity	100 kg per Folding Leaf (Up to 6 Leaves)	

FABRICATION STANDARDS & QUALITY COMPLIANCE

- Extrusion Tolerances: Conforms strictly to EN 12020-2 / DIN 17615 high precision architectural standard.
- Surface Coating: Qualicoat Certified Electrostatic Powder Coating (60-80 microns) / Qualanod Anodizing (15-20 microns).
- Gaskets: EPDM weatherseals according to DIN 7863. Hardware groove: Standard European groove compatibility.
- Quality Management: Extruded and assembled under certified ISO 9001:2015 industrial manufacturing systems.

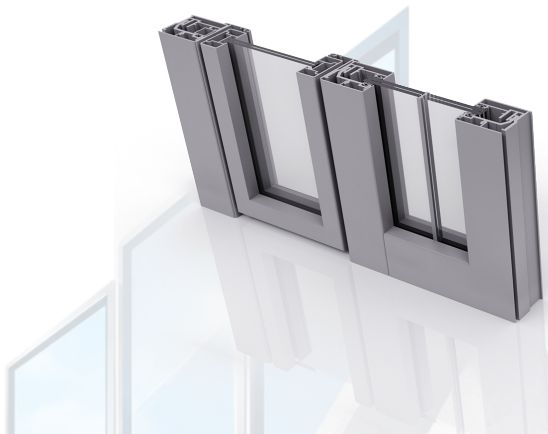
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IPA 45 - Acoustic Double-Glass Office Partition System

SERIES: IDA SERIES | SYSTEM CODE: IPA45 | EXTRUSION: EN AW-6063 T6

ENGINEERING 2D CAD PROFILE SCHEMATIC



MECHANICAL & ARCHITECTURAL SPECIFICATION MATRIX		MANUFACTURING STANDARD
Base Aluminum Alloy	EN AW-6063 T6 (Extrusion to EN 12020-2)	
Frame Architectural Depth	45 mm Perimeter Profiles / 100 mm Total Wall	
Sash / Vent Profile Depth	Flush Internal Glazing with Integrated Blinds	
Glazing Thickness Capacity	6 mm - 12 mm Double Acoustic Laminated Glass	
Thermal Barrier Insulation	Acoustic Cavity Isolation (Rw = 42 - 46 dB)	
Air Permeability (EN 12207)	Interior Commercial Specification	
Water Tightness (EN 12208)	Interior Commercial Specification	
Wind Load Resistance (EN 12210)	Interior Impact Safety Standard (BS 5234)	
Acoustic Sound Reduction	Rw = 42 - 46 dB Certified	
Maximum Sash Weight Capacity	80 kg Hinged Architectural Glass Door	

FABRICATION STANDARDS & QUALITY COMPLIANCE

- Extrusion Tolerances: Conforms strictly to EN 12020-2 / DIN 17615 high precision architectural standard.
- Surface Coating: Qualicoat Certified Electrostatic Powder Coating (60-80 microns) / Qualanod Anodizing (15-20 microns).
- Gaskets: EPDM weatherseals according to DIN 7863. Hardware groove: Standard European groove compatibility.
- Quality Management: Extruded and assembled under certified ISO 9001:2015 industrial manufacturing systems.

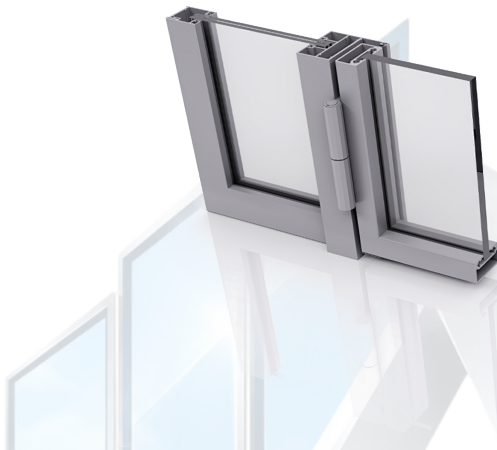
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IPA 30 - Minimalist Slimline Office Partition System

SERIES: IDA SERIES | SYSTEM CODE: IPA30 | EXTRUSION: EN AW-6063 T6

ENGINEERING 2D CAD PROFILE SCHEMATIC



MECHANICAL & ARCHITECTURAL SPECIFICATION MATRIX		MANUFACTURING STANDARD
Base Aluminum Alloy	EN AW-6063 T6 (Extrusion to EN 12020-2)	
Frame Architectural Depth	30 mm Ultra-Slimline Visible Profile	
Sash / Vent Profile Depth	Minimalist Micro-Channel Glazing	
Glazing Thickness Capacity	8 mm - 12 mm Monolithic Tempered Safety Glass	
Thermal Barrier Insulation	Acoustic Rubber Gasket Clamp Seals	
Air Permeability (EN 12207)	Interior Commercial Specification	
Water Tightness (EN 12208)	Interior Commercial Specification	
Wind Load Resistance (EN 12210)	Interior Non-Load Bearing Barrier	
Acoustic Sound Reduction	Rw = 34 - 38 dB	
Maximum Sash Weight Capacity	70 kg Glass Door Leaf	

FABRICATION STANDARDS & QUALITY COMPLIANCE

- Extrusion Tolerances: Conforms strictly to EN 12020-2 / DIN 17615 high precision architectural standard.
- Surface Coating: Qualicoat Certified Electrostatic Powder Coating (60-80 microns) / Qualanod Anodizing (15-20 microns).
- Gaskets: EPDM weatherseals according to DIN 7863. Hardware groove: Standard European groove compatibility.
- Quality Management: Extruded and assembled under certified ISO 9001:2015 industrial manufacturing systems.

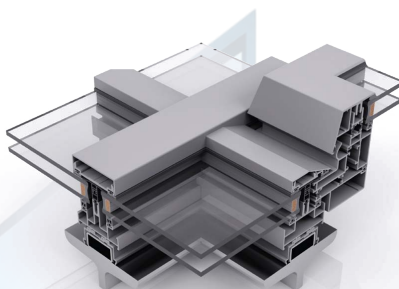
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SA 65 - Architectural Skylight & Atrium Roof System

SERIES: URAL SERIES | SYSTEM CODE: SA65 | EXTRUSION: EN AW-6063 T6

ENGINEERING 2D CAD PROFILE SCHEMATIC



MECHANICAL & ARCHITECTURAL SPECIFICATION MATRIX

MANUFACTURING STANDARD

Base Aluminum Alloy	EN AW-6063 T6 (Extrusion to EN 12020-2)
Frame Architectural Depth	65 mm Face Width / 120 mm Rafter Spine Depth
Sash / Vent Profile Depth	Exterior Clamping Pressure Plate & Cap
Glazing Thickness Capacity	24 mm - 44 mm Stepped Solar Control IGU
Thermal Barrier Insulation	Multi-Stage Internal EPDM Condensation Troughs
Air Permeability (EN 12207)	Class AE 1200 (EN 12152 / 1200 Pa)
Water Tightness (EN 12208)	Class RE 1500 (EN 12154 / 1500 Pa)
Wind Load Resistance (EN 12210)	High Structural Load / Snow Capacity (2400 Pa)
Acoustic Sound Reduction	Rw = 38 - 43 dB
Maximum Sash Weight Capacity	400 kg Glazing Node Capacity

FABRICATION STANDARDS & QUALITY COMPLIANCE

- Extrusion Tolerances: Conforms strictly to EN 12020-2 / DIN 17615 high precision architectural standard.
- Surface Coating: Qualicoat Certified Electrostatic Powder Coating (60-80 microns) / Qualanod Anodizing (15-20 microns).
- Gaskets: EPDM weatherseals according to DIN 7863. Hardware groove: Standard European groove compatibility.
- Quality Management: Extruded and assembled under certified ISO 9001:2015 industrial manufacturing systems.

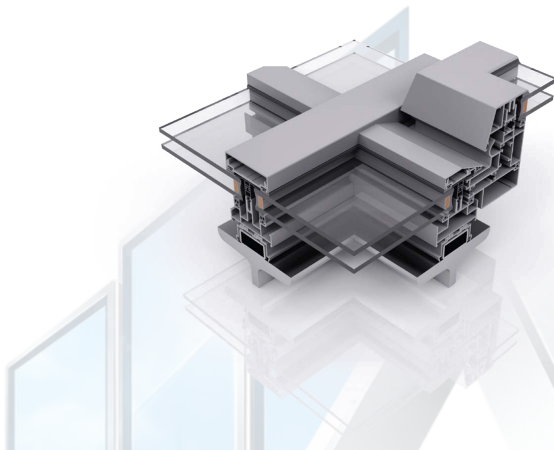
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SA 65-2 - Wintergarden Heavy-Span Steel Reinforced Roof

SERIES: URAL SERIES | SYSTEM CODE: SA65-2 | EXTRUSION: EN AW-6063 T6

ENGINEERING 2D CAD PROFILE SCHEMATIC



MECHANICAL & ARCHITECTURAL SPECIFICATION MATRIX

MANUFACTURING STANDARD

Base Aluminum Alloy	EN AW-6063 T6 with Internal Galvanized Steel
Frame Architectural Depth	65 mm Rafter with Internal Steel Tube (565-7101)
Sash / Vent Profile Depth	Reinforced Structural Glazing Rafter
Glazing Thickness Capacity	28 mm - 48 mm Triple Low-E Acoustic Glazing
Thermal Barrier Insulation	Continuous Polyamide + Drainage Gasket Matrix
Air Permeability (EN 12207)	Class AE 1200 (EN 12152 / 1200 Pa)
Water Tightness (EN 12208)	Class RE 1800 (EN 12154 / 1800 Pa)
Wind Load Resistance (EN 12210)	High Wind & Snow Load Capacity (3200 Pa)
Acoustic Sound Reduction	Rw = 40 - 45 dB
Maximum Sash Weight Capacity	550 kg Reinforced Span Capacity

FABRICATION STANDARDS & QUALITY COMPLIANCE

- Extrusion Tolerances: Conforms strictly to EN 12020-2 / DIN 17615 high precision architectural standard.
- Surface Coating: Qualicoat Certified Electrostatic Powder Coating (60-80 microns) / Qualanod Anodizing (15-20 microns).
- Gaskets: EPDM weatherseals according to DIN 7863. Hardware groove: Standard European groove compatibility.
- Quality Management: Extruded and assembled under certified ISO 9001:2015 industrial manufacturing systems.

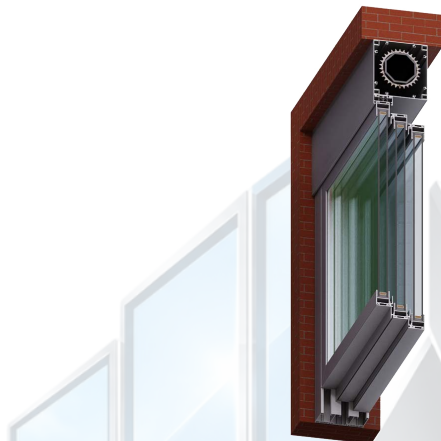
FACTORY DIRECTIVE: Official technical specifications verified for tender submissions, fabrication, and structural installation.

DIRECT CONTACT: +218 92 429 5050 | +218 91 614 1616 | Tripoli Industrial Complex, Libya | www.ainzara.ly

GSA 130 - Motorized Guillotine Vertical Sliding System

SERIES: LOGAN SERIES | SYSTEM CODE: GSA130 | EXTRUSION: EN AW-6063 T6

ENGINEERING 2D CAD PROFILE SCHEMATIC



MECHANICAL & ARCHITECTURAL SPECIFICATION MATRIX		MANUFACTURING STANDARD
Base Aluminum Alloy	EN AW-6063 T6 (Extrusion to EN 12020-2)	
Frame Architectural Depth	130 mm Vertical Guide Jamb Channel	
Sash / Vent Profile Depth	3-Panel Cascading Motorized Telescopic Sashes	
Glazing Thickness Capacity	8 mm - 10 mm Tempered / 20 mm IGU Double Glass	
Thermal Barrier Insulation	High-Density Brush & Acoustic Felt Seals	
Air Permeability (EN 12207)	Automated Retractable Weather Gaskets	
Water Tightness (EN 12208)	Class 7A (EN 12208 / 300 Pa)	
Wind Load Resistance (EN 12210)	Class C3 (EN 12210 / 1200 Pa)	
Acoustic Sound Reduction	Rw = 32 - 37 dB	
Maximum Sash Weight Capacity	220 kg Automated Belt Drive Motor System	

FABRICATION STANDARDS & QUALITY COMPLIANCE

- Extrusion Tolerances: Conforms strictly to EN 12020-2 / DIN 17615 high precision architectural standard.
- Surface Coating: Qualicoat Certified Electrostatic Powder Coating (60-80 microns) / Qualanod Anodizing (15-20 microns).
- Gaskets: EPDM weatherseals according to DIN 7863. Hardware groove: Standard European groove compatibility.
- Quality Management: Extruded and assembled under certified ISO 9001:2015 industrial manufacturing systems.

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AUTOMATED GLASS PROCESSING PLANT (TRIPOLI FACILITY)

AinZara operates modern CNC glass cutting, automated double-edge grinding, convection tempering, and robotic insulating glass assembly lines for high-performance architectural facades across Libya.

CERTIFIED QUALITY STANDARDS: EN 1279-2 (IGU Gas Retention), EN 12150-1 (Tempered Safety), EN 14449 (Laminated).

GLAZING MAKEUP SPECIFICATION	Ug VALUE (W/m2K)	SHGC	ACOUSTIC Rw	RECOMMENDED
6mm Clear Monolithic Toughened	5.7 W/m2K	0.82	31 dB	Interior Partition (IPA 30)
8mm / 10mm Clear Tempered Safety	5.6 W/m2K	0.79	33 dB	Guillotine Balcony (GSA 130)
6mm + 12mm Air + 6mm Clear DGU	2.8 W/m2K	0.72	34 dB	Economic Casement (WA 45)
6mm + 16mm Argon + 6mm Low-E IGU	1.1 W/m2K	0.40	37 dB	Standard Thermal (WAT 63)
6mm Solar + 16Ar + 6mm Low-E IGU	1.0 W/m2K	0.28	38 dB	Curtain Wall Facade (CWA 50)
8mm + 20Ar + 8mm Heavy Acoustic IGU	1.1 W/m2K	0.38	42 dB	Lift & Slide Door (SAT 120)
4.4.2 PVB Laminated + 16Ar + 6 Low-E	1.1 W/m2K	0.36	44 dB	Security & Acoustic Glazing
6.6.2 Acoustic PVB Laminated	5.2 W/m2K	0.68	46 dB	Office Partition (IPA 45)
6 Low-E + 14Ar + 4 Float + 14Ar + 6 Low-E	0.6 W/m2K	0.29	45 dB	Passive House Triple IGU

PROCESSING EQUIPMENT & MAXIMUM CAPACITIES

- Max Tempered Sheet Size: 2440 x 5000 mm (Glass thicknesses from 4mm up to 19mm).
- Double Glazing Automatic Sealing: Automated two-component polysulfide or structural silicone edge sealing.
- High-Altitude / Pressure Equalization: Capillary breather tubes installed for projects in mountainous areas.
- Acoustic Lamination Autoclave: Multi-layer acoustic PVB & SentryGlas (SG) ionoplast interlayers for cyclone/bullet resistance.
- Spandrel & Ceramic Fritted Glass: High-durability screen-printed ceramic enamel baked into glass during tempering.

Ain Zara
Glass Processing & Aluminum Manufacturing

TRIPOLI INDUSTRIAL HEADQUARTERS

Ain Zara Industrial District, Tripoli, Libya - Official Factory Complex

DIRECT FACTORY INQUIRY DIRECTORY & TENDER DESKS

DIRECT FACTORY TELECOMMUNICATIONS & WHATSAPP

PRIORITY LINE 01

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DIGITAL ARCHITECTURAL PORTAL & ONLINE TOOLS

Visit the official AinZara web platform for interactive 3D WebGL models, CAD DWG files, open-IFC BIM components for Autodesk Revit, and the automated cutting list calculator.

WEBSITE: <https://ainzara.ly> | **EMAIL:** info@ainzara.ly

Operating Hours: Saturday to Thursday, 08:30 to 18:00 (Tripoli Time)

MANUFACTURING ACCREDITATIONS & QUALITY SEALS:

ISO 9001:2015 Quality Management Certified * CE Mark Declaration of Performance

Qualicoat Standard No. 42918 * Qualanod Architectural Anodizing Standard No. 1824

EN AW-6063 T6 Alloy Extrusion strictly conforming to DIN 17615 / EN 12020-2